

URBAN CITY CLEAN (MISSION CLEAN CITY)

Smart Digital Cleanliness Monitoring System for Smart Bengal

Strategic Project Proposal & Scalability Roadmap

Prepared For: Urban Development Department, Government of West Bengal
Current Phase: Live Prototype (Birnagar Municipality)
Future Scope: Multi-Municipality State-Wide Ecosystem
Date: May 2026

1. Project Overview

Urban City Clean is a smart digital platform designed to improve city cleanliness and waste management across urban areas of West Bengal. The system helps citizens directly report issues such as:

- **Garbage dumping**
- **Dirty roads**
- **Drainage blockage**
- **Unclean public areas**
- **Overflowing dustbins**
- **Improper waste disposal**
- **Streetlight issues**
- **Waterlogging**
- **Illegal dumping**
- **Lack of sanitation facilities**
- **Public health concerns related to cleanliness**
- **Other cleanliness-related issues**

Through a user-friendly website or mobile platform, citizens can easily upload images and use live location tracking to report problems. These complaints are instantly sent to municipal authorities for faster action and monitoring.

*[!NOTE] **Technical Context:** This platform is currently operating as a **fully functional prototype** (Mission Clean City) specifically tailored for the Birnagar Municipality. It acts as a centralized, AI-driven Civic Technology platform designed to modernize municipal grievance redressal and replace traditional, disconnected physical ledgers.*

2. Main Objectives

- **Improve Urban Cleanliness:** Direct community involvement in maintaining hygiene.
- **Faster Complaint Management:** Transition from paper-based and fragmented complaint systems to a unified digital portal.
- **Increase Citizen Participation:** Make reporting issues as simple as taking a photo.
- **Reduce Manual Workload:** Equip administrators with automated dashboards and AI to handle data easily.
- **Support Smart Bengal:** Align with state-wide Smart City and digital governance initiatives.

3. How the System Works

The platform ensures a seamless flow of information between the public and the cleaning staff. Below is the step-by-step visual workflow of the system in action.

3.1 Citizen Side Workflow

1. **Secure Login:** Citizens access their accounts using their registered mobile number and password.

The screenshot shows the 'Citizen Login' page for Birnagar Municipality. The header includes the municipality logo, name, and navigation links: Home, About, Suggestions, Login, and a 'Report Issue' button. The main content area features a 'Citizen Login' title with a link to 'Or create a new account'. Below this is a login form with fields for 'Mobile Number' (containing '988') and 'Password' (masked with dots). A 'Forgot your password?' link is positioned below the password field. A 'Sign in' button is at the bottom of the form. A chat icon is visible in the bottom right corner.

2. **Tracking Dashboard:** After login, citizens can see the live status of all their reported issues.

The screenshot shows the 'My Dashboard' page for a logged-in citizen, Roni Paul. The header includes the municipality logo, name, and navigation links: Home, About, Suggestions, Submit Issue, and 'My Dashboard'. The user's name 'Hi, Roni Paul' and a 'Logout' button are in the top right. The dashboard features a profile section with a profile picture, name 'Roni Paul', ID 'BM-199817', and a QR code. Below the profile are four cards displaying: 'MOBILE NUMBER 9883407189', 'AADHAAR NUMBER XXXX-XXXX-1704', 'WARD NUMBER 5', and 'TOTAL REPORTS 7'. An 'Edit Profile' button is next to the QR code. Below the profile section is a 'Track an Issue' section with a search bar labeled 'Enter Problem ID (e.g., BIR-WARD4-1029)' and a 'Search' button. At the bottom is the 'My Submission History' section, which has two tabs: 'Issues Reported' (with a count of 7) and 'Submitted Suggestions' (with a count of 1). Below the tabs are two dropdown menus for 'Status' (set to 'All Statuses') and 'Category' (set to 'All Categories').

3. **Reporting an Issue:** A simple form allows citizens to upload photos and descriptions, with GPS location captured automatically.



Birnagar Municipality

MISSION CLEAN CITY

Home

About

Suggestions

Submit Issue

My Dashboard



Hi, Roni Paul

Logout

Report a Problem



Your Ward: 5

Submission Limits

You can submit max 2 reports daily and 10 per month.

Daily Left: 2/2

Monthly Left: 10/10

Photos of the Issue (Max 3, Min 1) *

JPEG/JPG Images only. Photos will be auto-compressed to under 75KB.

Upload Photo 1 *

Upload Photo 2

Upload Photo 3

Video Proof (Optional, Max 100MB)

Upload Video (MP4, AVI, etc.)

Problem Category *

Select a Problem Category

Pin Exact Location (Required)

Click on the map to place a pin at the exact location of the problem. Address will be fetched automatically.



No location selected.

Clear Pin

Fetch Address *

Address will auto-fill when you pin the location

Submit Issue

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4. **Providing Suggestions:** Citizens can participate in city planning by submitting suggestions for cleanliness improvements.



Birnagar Municipality
MISSION CLEAN CITY

[Home](#) [About](#) [Suggestions](#) [Submit Issue](#) [My Dashboard](#) [Hi, Roni Paul](#) [Logout](#)

Suggestions Box

Have an idea on what we can improve for our city? We're all ears.

Submission Limits
You can submit max 1 suggestion daily and 5 per month.

Daily Left: 1/1
Monthly Left: 5/5

Your Information

Full Name
Roni Paul

Mobile Number
9883407189

Aadhaar Number
XXXX-XXXX-1704

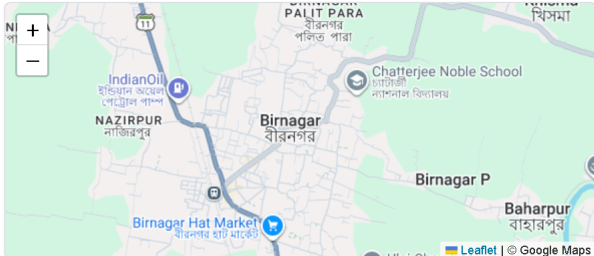
Your Ward Number
Ward 5

Your Address
Birnagar, Nadia, West Bengal, India - Updated

Improvement Area

Target Ward Number *
Select Target Ward

Target Address *
Full address of the area that needs improvement

Pin Exact Location (Optional but recommended)
Click on the map to place a pin at the exact location that needs improvement.


No location selected. [Clear Pin](#)

Suggestion Category *
Select a Category

Your Idea / Suggestion
Tell us how we can make Birnagar better...

Photo of this place *

Choose File No file chosen

JPEG//PG images only. Auto-compressed to under 75KB.

Submit Suggestion

Submissions are reviewed weekly by the municipal board.

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3.2 Administrative Side Workflow

1. **Admin Portal Login:** Municipal staff use their hardened, non-editable Employee IDs to log in.



Birnagar Municipality
MISSION CLEAN CITY

HomeAboutSuggestionsLogin ▾

Report Issue

Admin Portal Login

Authorized Personnel Only

Employee ID

adrr

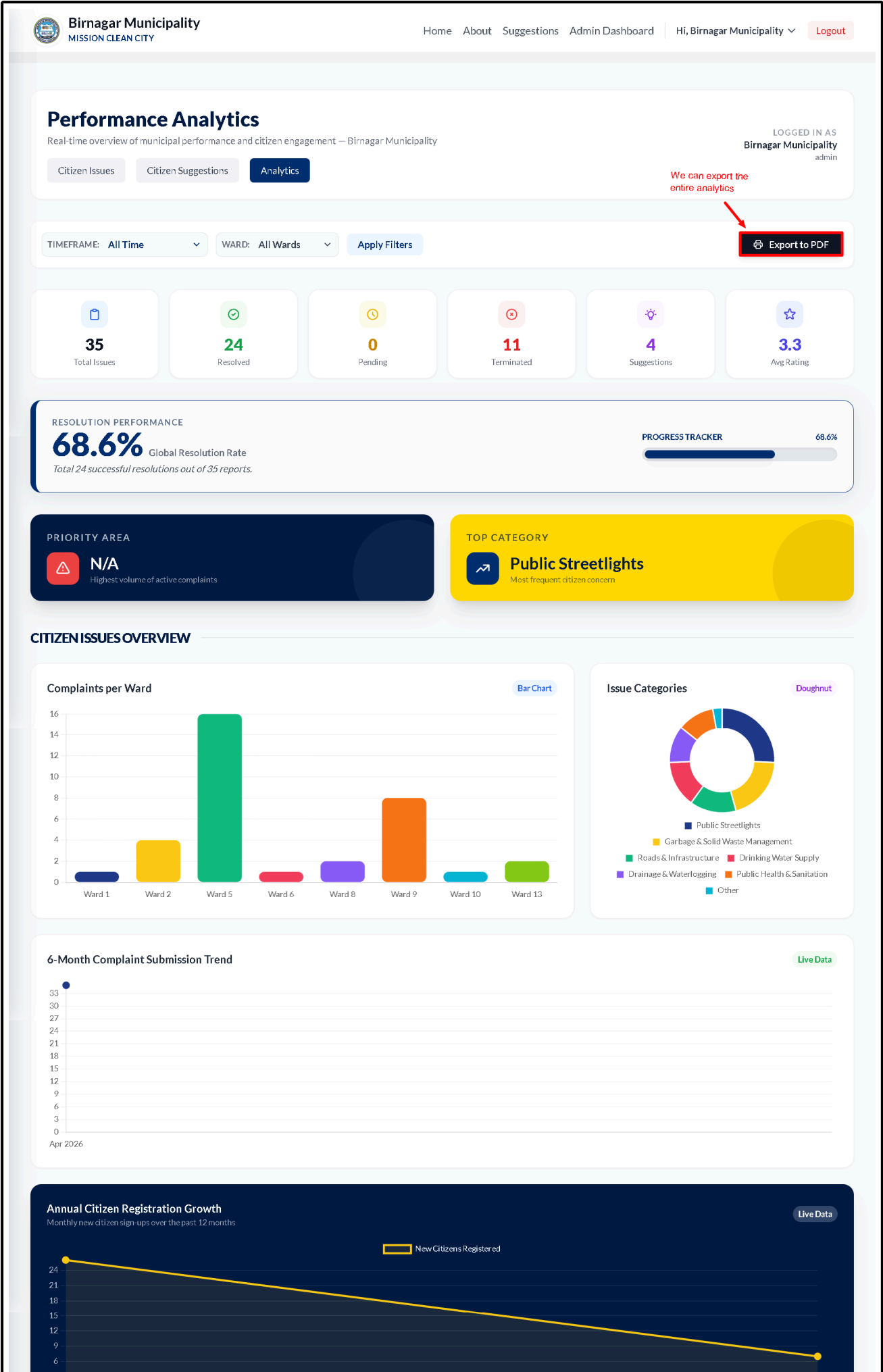
Password

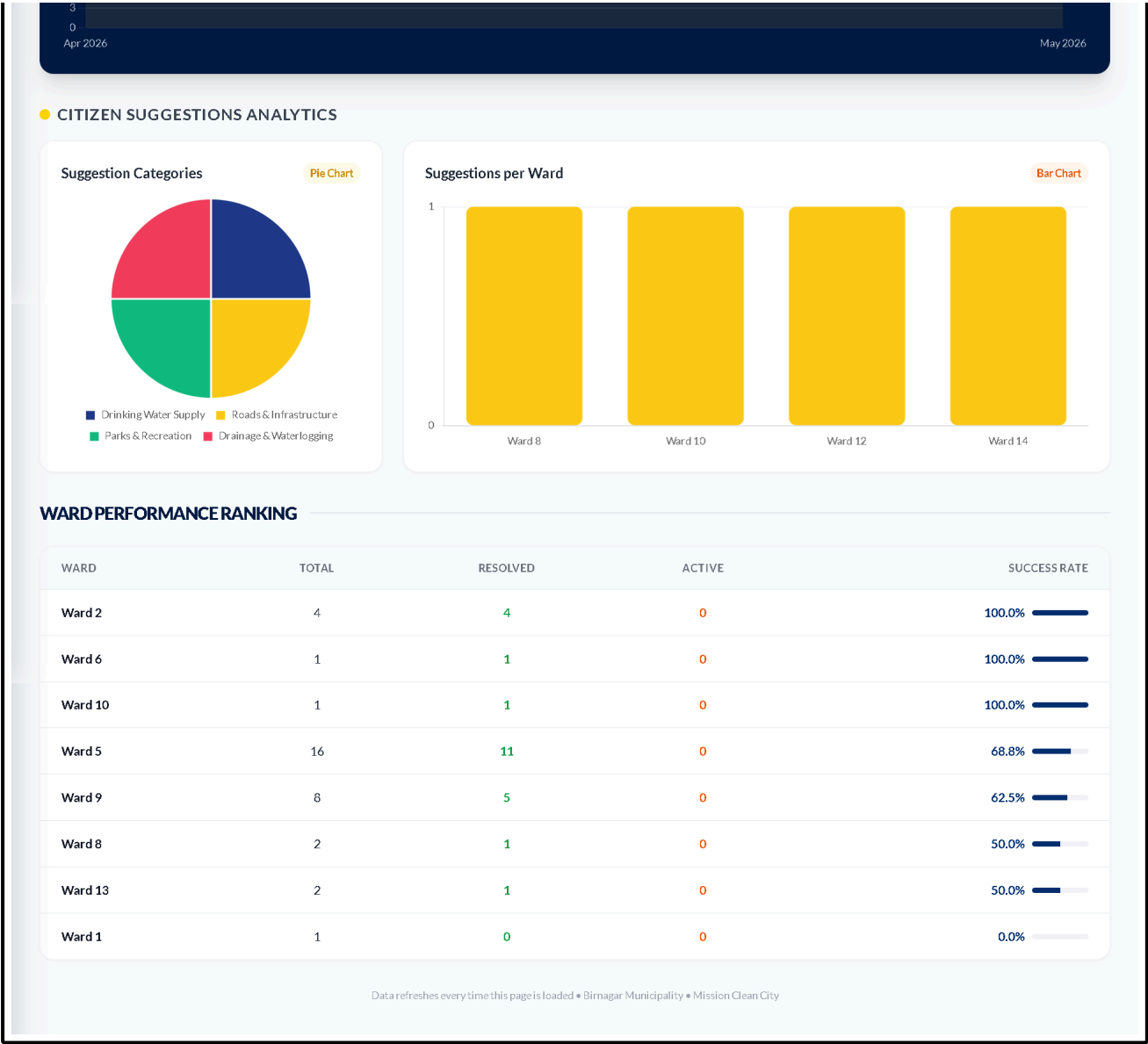
.....

Sign In to Dashboard

Forgot your admin password?

2. **Command Center Dashboard & Analytics:** The main dashboard provides real-time KPIs and ward-wise analytics.







Birnagar Municipality

MISSION CLEAN CITY

Home

About

Suggestions

Admin Dashboard

Hi, Birnagar Municipality

Logout

Admin Dashboard

Welcome, Birnagar Municipality (ID: admin)

Search ID, Name, Mobile, Description, Address...

All Wards

All Statuses

All Categories

All Priorities

From: dd-mm-yyyyTo: dd-mm-yyyy

Search

QUICK FILTERS:

Urgent & Pending

In Progress

Unrated Resolved

Appealed

Today's Issues

Pending0

In Progress0

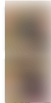
Solved24

Terminated11

Citizen Issues

PAGE 1 OF 4

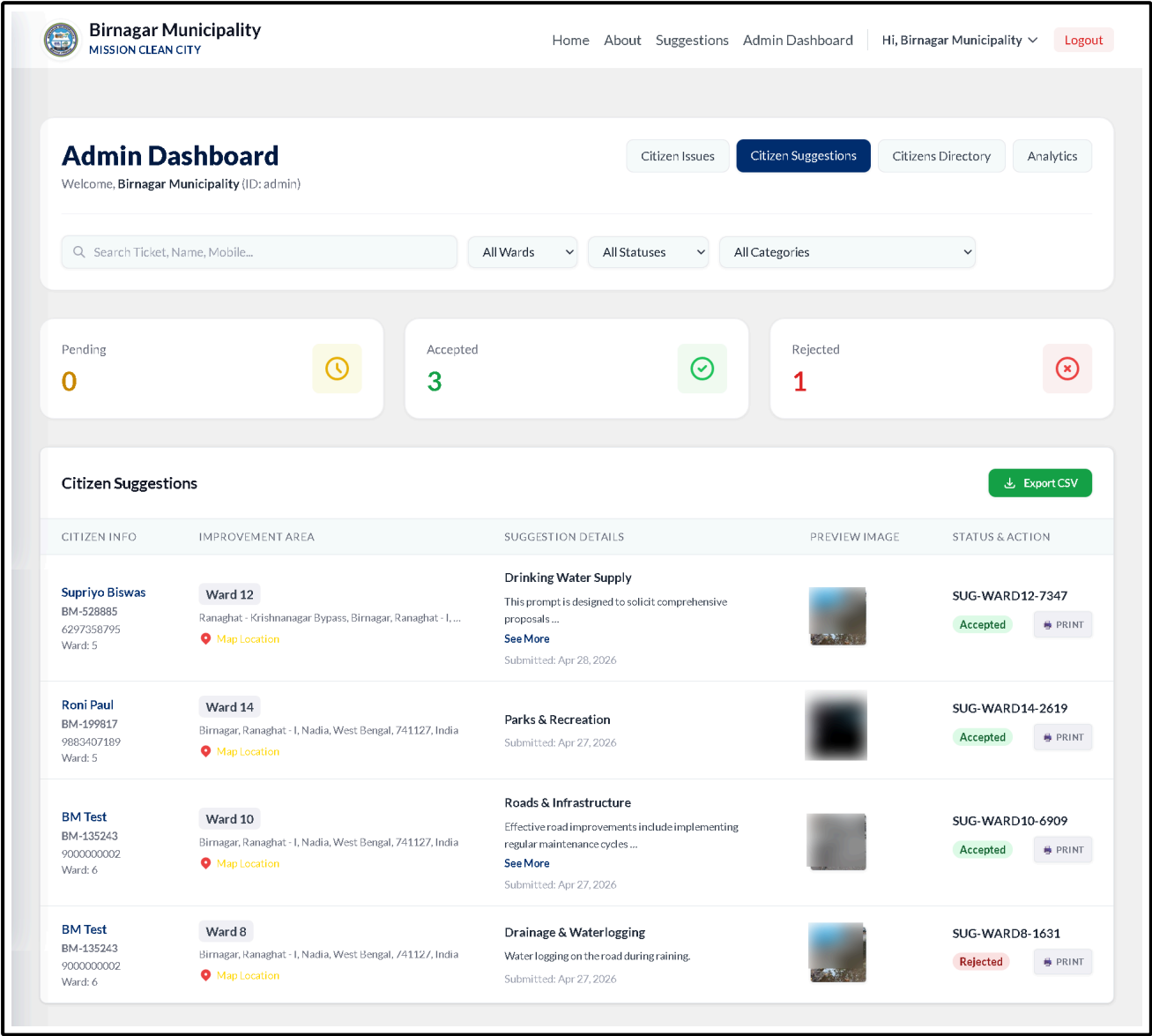
Export CSV

☐	CITIZEN INFO	LOCATION & ID	DETAILS	PROOF	RATING	ACTION & STATUS
☐	Hello D BM-120539 9313845976 Ward: 2	BIR-WARD2-8579 Birnagar, Ranaghat - I, Nadia, West Bengal, 741127... Map Location	Public Streetlights Streetlight pole is physically damaged, tilted, or unsafe Appeal Reason: Please look at this matter more carefully. SUBMITTED: APR 30, 2026		★★★★☆ Rated	Resolved Done. Status Locked PRINT
☐	Roni Paul BM-199817 9883407189 Ward: 5	BIR-WARD5-7745 Birnagar, Ranaghat - I, Nadia, West Bengal, 741127... Map Location	Drinking Water Supply Appeal Reason: Please take it back. SUBMITTED: APR 30, 2026		Not Rated	Terminated Already done with the photo. Status Locked PRINT
☐	Roni Paul BM-199817 9883407189 Ward: 5	BIR-WARD5-4630 Birnagar, Ranaghat - I, Nadia, West Bengal, 741127... Map Location	Drinking Water Supply SUBMITTED: APR 30, 2026		★★★★☆ Rated	High Resolved Great Work. Status Locked PRINT
☐	Hello E BM-783700 7143761132 Ward: 9	BIR-WARD9-9280 Birnagar, Ranaghat - I, Nadia, West Bengal, 741127... Map Location	Public Streetlights Streetlight bulb is blown out or not turning on at night SUBMITTED: APR 29, 2026		★★★★☆ Rated	Resolved Status Locked PRINT
☐	Hello D BM-120539 9313845976 Ward: 2	BIR-WARD2-9705 Ranaghat - Krishnanagar Bypass, Birnagar, Ranagh... Map Location	Public Streetlights Appeal Reason: It is a serious issue for our citizens. Please give more priority in this matter. SUBMITTED: APR 29, 2026		★★★★☆ Rated	Medium Resolved Status Locked PRINT
☐	Hello C BM-386461 8264415597 Ward: 8	BIR-WARD8-4768 Birnagar, Ranaghat - I, Nadia, West Bengal, 741127... Map Location	Drinking Water Supply SUBMITTED: APR 29, 2026		★★★★☆ Rated	Resolved Status Locked PRINT
☐	Hello C BM-386461 8264415597 Ward: 8	BIR-WARD8-9353 Birnagar, Ranaghat - I, Nadia, West Bengal, 741127... Map Location	Drainage & Waterlogging Appeal Reason: Accept SUBMITTED: APR 29, 2026		Not Rated	Terminated Not Possible. Status Locked PRINT
☐	Hello B BM-470021 9175972450 Ward: 9	BIR-WARD9-3217 Birnagar, Ranaghat - I, Nadia, West Bengal, 741127... Map Location	Public Streetlights Appeal Reason: Yes SUBMITTED: APR 29, 2026		★★★★☆ Rated	Resolved Status Locked PRINT
☐	Hello B BM-470021 9175972450 Ward: 9	BIR-WARD9-7294 Birnagar, Ranaghat - I, Nadia, West Bengal, 741127... Map Location	Garbage & Solid Waste Management Appeal Reason: Yes		Not Rated	Terminated Not acceptable Status Locked PRINT

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4. Reviewing Citizen Suggestions: A dedicated section for administrators to evaluate community-driven ideas.



4. Key Features & Technical Implementation

The system successfully bridges the communication gap through a dual-ecosystem tailored for both the public and the administration.

4.1 Citizen Side (Public Interface)

- **Online Complaint Submission & Photo Upload:** Easy-to-use multimedia reporting.
- **GPS Location Tracking:** Pinpoints the exact location of the issue.
- **Live Complaint Status:** Citizens can track their complaints (Pending → In Progress → Resolved) in real-time.

- **Secure Registration:** OTP-based mobile/email verification ensures all users are genuine residents.

4.2 Administration Side (Command Center)

- **Smart Dashboard:** Visual KPIs displaying total complaints, resolution rates, and ward-wise performance.
- **Complaint & Worker Monitoring:** Ability to manage tasks and generate printable PDF work orders for on-ground staff directly from the portal.
- **Area-Wise Cleanliness Tracking:** Identifies infrastructure "hotspots" requiring urgent attention.
- **Hardened Identity Protection:** Administrative identities (Employee ID and Full Name) are locked to prevent internal impersonation. Admins manage profiles via OTP-verified secure workflows.

4.3 AI Civic Assistant (Birni AI)

- **24/7 Chatbot:** Integrated AI assistant (powered by Google Gemini) capable of answering FAQs and guiding users on municipal policies.
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5. Technical Architecture & Security Framework

For Technical Review Members: The platform is built on modern, secure, and scalable technologies capable of handling state-wide traffic.

- **Backend Framework:** Python / Django (Highly secure and scalable, capable of supporting multi-tenant databases).
- **AI Integration:** Google Gemini NLP models.
- **Hardened Authentication:** Multi-factor authentication (MFA) via OTP for all critical account modifications.
- **Identity Protection:** Sensitive citizen data (such as Aadhaar numbers) is encrypted at rest using Advanced Symmetric Encryption (Fernet).
- **Automated Credential Dispatch:** Credentials for administrative staff are dispatched via encrypted official email, ensuring strict Role-Based Access Control (RBAC).

5.1 Key Security & Administrative Workflows

- **Admin Identity Hardening:** Every Administrator is assigned a unique **Employee ID** at registration. This ID and their **Full Name** are locked at the database level and cannot be modified by the user, preventing internal identity fraud.
 - **Secure Email Modification:** Any change to an administrative email address requires mandatory 6-digit OTP verification sent to the *new* email address before the change is committed.
 - **Isolated Password Recovery:** The system maintains separate, cryptographically secure recovery pipelines for Citizens and Administrators to prevent cross-account attacks.
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6. Benefits of the Project

- **Faster response** to public complaints.
 - **Cleaner urban environment** and healthier communities.
 - **Transparent monitoring system** that builds confidence in local government.
 - **Better communication** between citizens and municipality staff.
 - **Optimized Budgeting:** Analytics allow municipalities to deploy funds exactly where infrastructure is failing.
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7. Future Scope & Scalability

The architecture of the prototype was engineered to facilitate rapid expansion to build a massive state-wide infrastructure.

- **State-Wide Monitoring System (SaaS Model):** A single platform capable of hosting multiple municipalities simultaneously (e.g., Ranaghat, Krishnanagar) with complete data isolation for each.
 - **Mobile Application:** Transitioning from a web portal to dedicated, native **Android and iOS applications** with offline-mode capabilities.
 - **IoT Smart Dustbin Integration:** Future compatibility to connect with physical smart bins that automatically alert the municipality when full.
 - **Advanced Admin Workstations:** The Admin portal will remain strictly optimized for Laptop/Desktop environments to efficiently manage GIS mapping and large datasets.
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8. Conclusion

Urban City Clean (Mission Clean City) is a modern technology-based initiative aimed at building a cleaner, smarter, and healthier West Bengal. This platform can help municipalities improve waste management efficiency while encouraging active citizen participation in maintaining city cleanliness. By transitioning this prototype into a Multi-Municipality platform, the Urban Development Department has the opportunity to establish a new gold standard for proactive civic administration.

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